Modelling Particle Production in Analog (3+1) Inflationary Cosmologies Using High-Performance Computing

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Introduction - Analog Universes

1. Inflation created particles

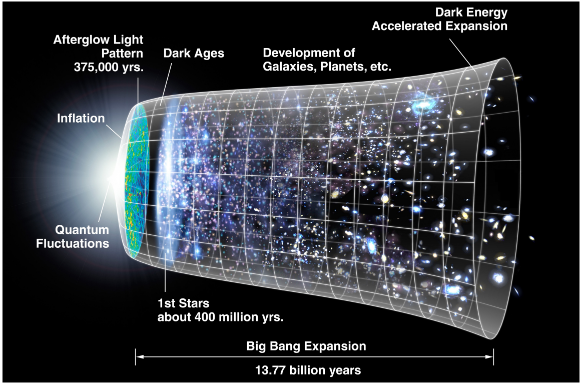
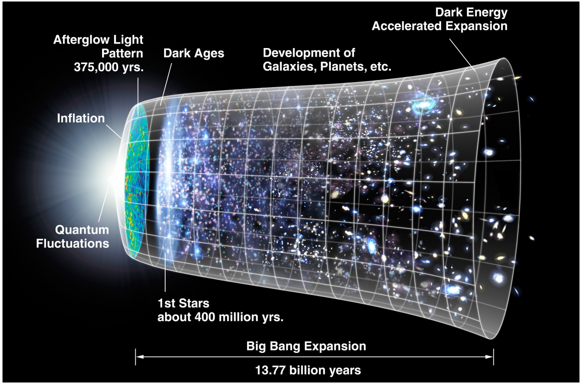


Figure: History of the universe highlighting period of rapid expansion (Source)

Introduction - Analog Universes

1. Inflation created particles
2. Cannot directly observe these particles



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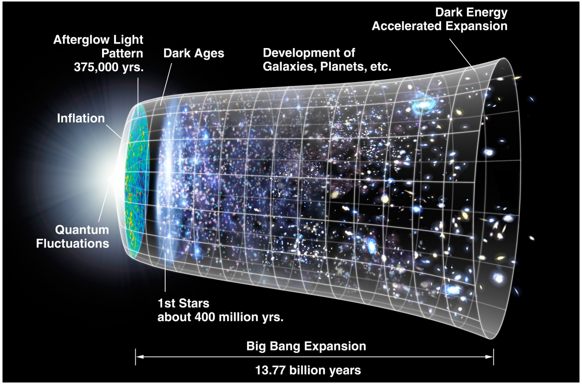

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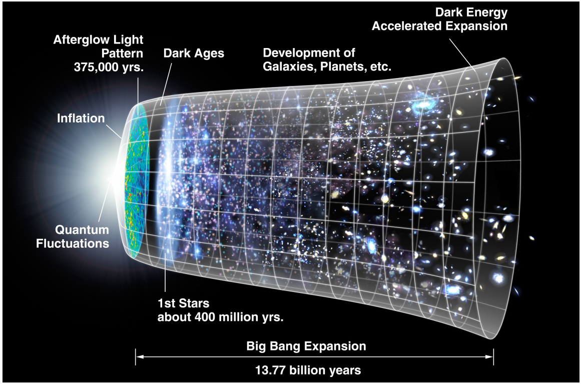
4. Can make gas in the lab $\implies$ can *effectively* observe inflation particles



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1. Inflation created particles
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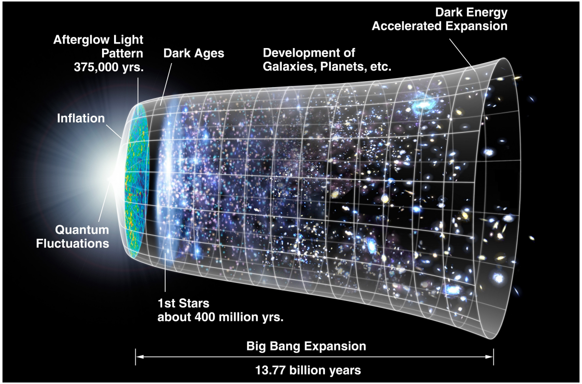
5. Lab based system $=$ "analog universe"



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Introduction - The Story So Far

- Many investigations into analog universes

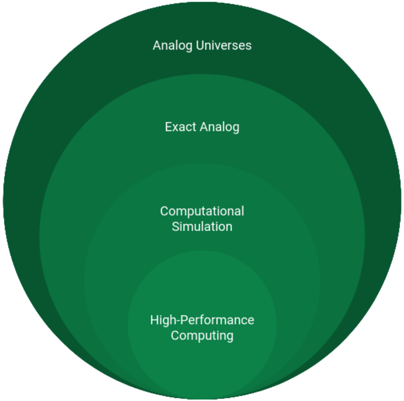

Figure: Subsets of increasingly specific literature

Introduction - The Story So Far

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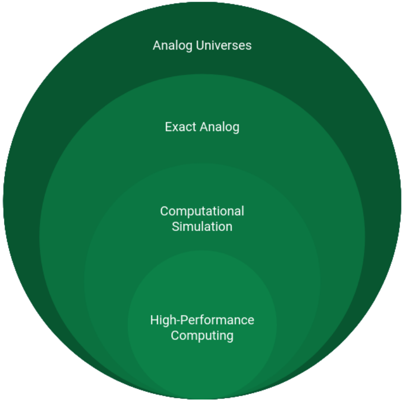

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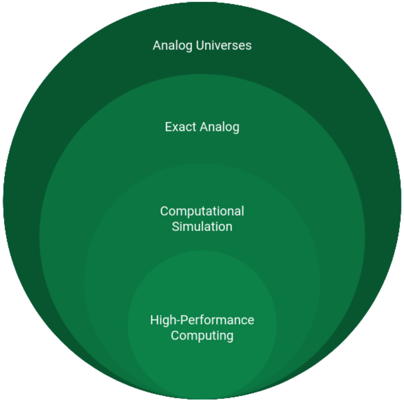

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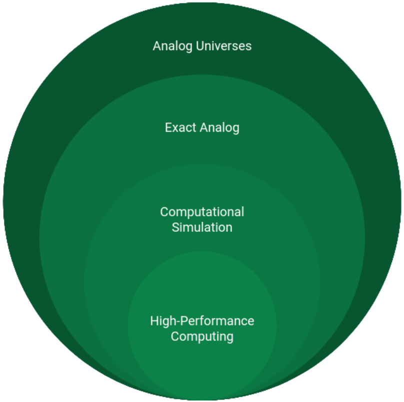
- Many investigations into analog universes
- Few into exact analogs
- Fewer into computational simulation
- Even fewer using high-performance computing



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Methods - Physical

Cosmological

1. Spacetime
   - increase in spatial-volume (inflation)
   - constant speed of light
2. Observer dependent quantum field excitations
3. Spontaneous particle production

Inflation determines particle momentum

Condensed Matter

1. Bose-Einstein condensate
   - fixed volume
   - decreasing speed of sound
2. Oscillatory phase perturbations
3.

Phase perturbations are damped oscillators

Methods - Mathematical

- One equation describes both ultra-cold quantum gases and inflation particles
- One equation for each phononic k
- Universe expansion starts and stops at finite times

The Field Equation

$$\partial_t^2 \widetilde{\theta}_{\mathbf{k}} - \frac{\dot{a}}{a} \partial_t \widetilde{\theta}_{\mathbf{k}} - a c_0^2 k^2 \widetilde{\theta}_{\mathbf{k}} = 0$$

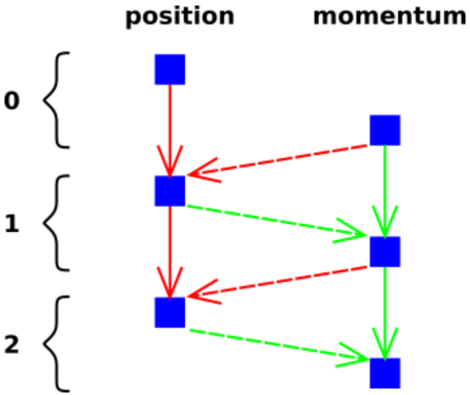
$$\widetilde{\theta}_{\mathbf{k}}(0) = \widetilde{\theta}_{\mathbf{k}0} \quad \partial_t \widetilde{\theta}_{\mathbf{k}}(0) = -U_0 n_{\mathbf{k}0}/\hbar$$

Methods - Computational

- Symplectic integrator e.g. leapfrog to minimize energy drift
- Parareal for parallel-in-time i.e. predictor-corrector
- GPU
- Distributed on CWU's supercomputer with 12 P100 GPUs

Methods - Solving The Field Equation

- Need to try to conserve energy



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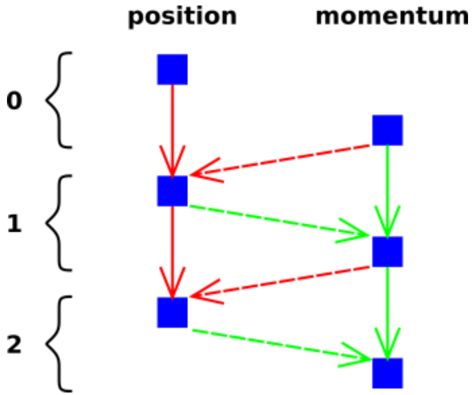

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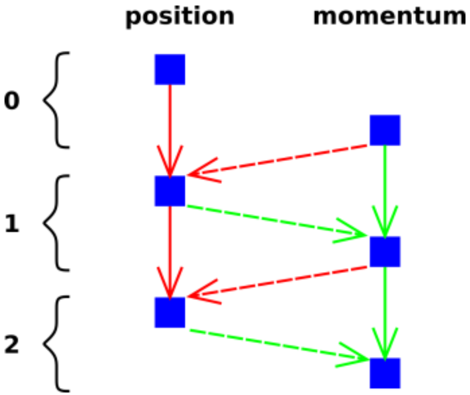

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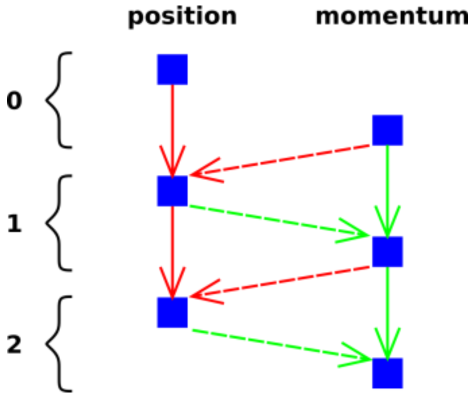
Methods - Solving The Field Equation

- Need to try to conserve energy
- Use *symplectic integrator*
- Use new momentum to calculate position
- More complex but more stable



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Methods - High-Performance Computing

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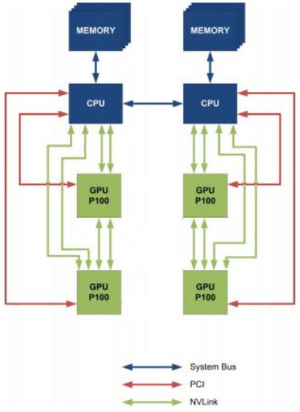

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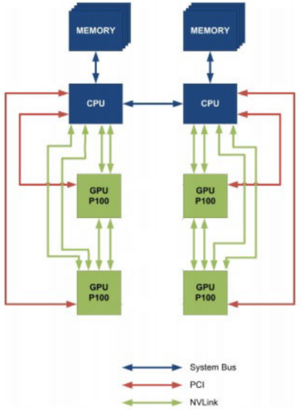

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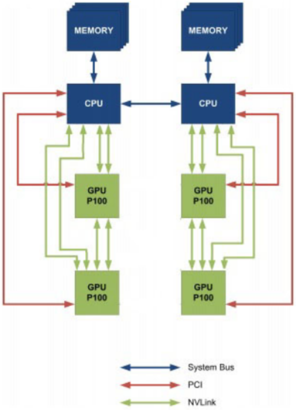

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Methods - High-Performance Computing

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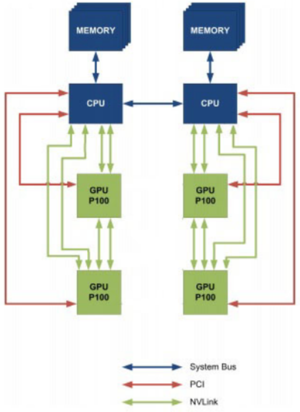
- Use CWU's "Turing" supercomputer
- 4x4 NVIDA P100 GPUs

  - Each 83% performance of modern RTX 3060



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Analytic Results - Simple Expansion

- Simple expansion: $a(t) = e^{-t/t_s}$

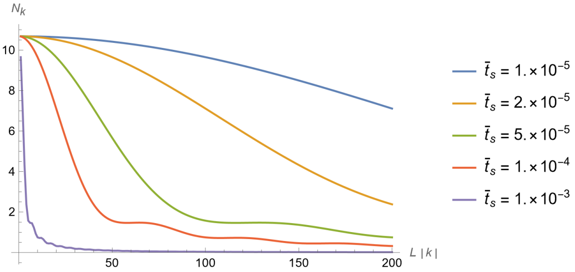

Figure: The momentum distribution when the universe stops expanding for multiple effective rates of expansion

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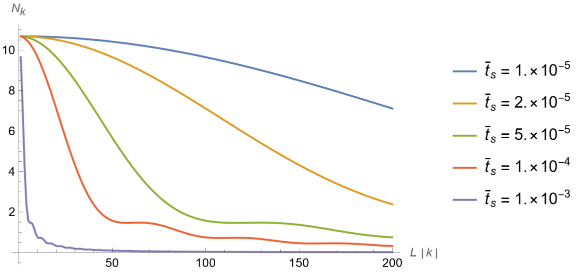

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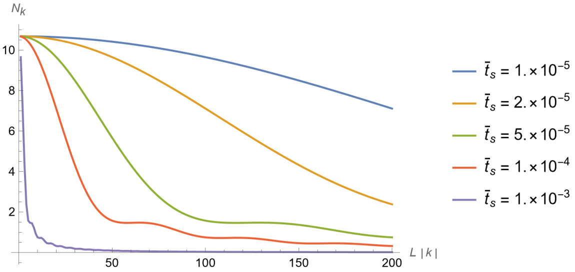

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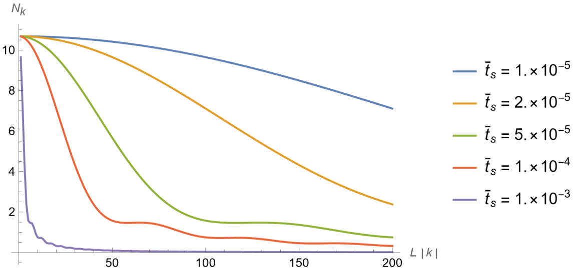

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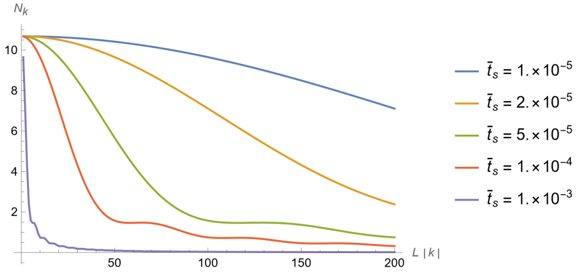
- Population drops off quickly with k



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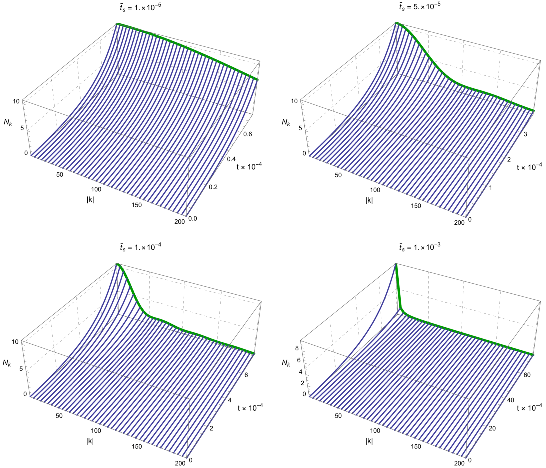

Figure: Momentum distribution throughout the expansion

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- There are many equations that need to be solved with special methods to conserve energy
- Use high-performance techniques and hardware for reasonable execution time
- Expect the distribution of momenta to depend on the length of the expansion
- Expect the population to be high for low k and drop off quickly

Acknowledgements

- Dr. Andy Piacsek (CWU, Physics)
- Dr. Szilard VAJDA (CWU, Computer Science)
- Dr. Micheal Braunstein (CWU, Physics)
- Dr. Brandon Peden (WWU, Physics & Astronomy)

Thank you & Code source

Thank you for your attention.

The code for this project is available at:

- GitHub (QR code): `github.com/nondairyneutrino/thesis`
- My website: `nathanwchapman.com`